Sara Nishiyama
Makayla Byas
Vanessa Gonzales
DA6813
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Bank Marketing Case Study

I. Executive Summary

The following case study addresses the challenge of optimizing telemarketing campaigns for a European retail banking institution. The primary objective was to predict customer subscriptions to long-term deposits. This task was complicated by a significant class imbalance where only 11.3% of the 41,188 observations resulted in a successful "yes". With a data driven approach that maximizes conversion rates while minimizing operational costs, the bank can transition from a broad, inefficient "random calling" strategy.

After extensive feature engineering, we implemented and compared a Logistic Regression model and a Linear Discriminant Analysis model. Furthermore, a "down-sampling" technique was used to prevent model bias toward the majority of "no" classes. Performance of the models was inspected using the Area Under the ROC Curve and a 5-Fold cross validation framework. Following the comparison, the Logistic Regression model was preferred because of its slightly greater ability to correctly filter out non-subscribers achieving a specificity score of 78.58%. This is critical for preserving the bank's resource allocation.

The analysis of the predictive model revealed that subscription behavior is driven by macroeconomic climate, contact method, and existing relationships. The 3-month Euribor rate and Consumer Confidence index were top predictors, indicating that the bank's sales windows are heavily influenced by a broader economic environment. Successful previous outcomes and landline communication were also strong indicators of future success. Specialized outreach to establish leads should be prioritized especially when the economic environment is in their favor.

II. Problem

The task for this study is to develop predictive models to optimize telemarketing campaigns for a mid-sized European retail bank promoting long-term deposit accounts. Currently, the bank invests considerable resources in phone calls, emails, and in-branch outreach,

yet only a small percentage of customers subscribe. We aim to build and compare Logistic Regression and Linear Discriminant Analysis models to identify which customers are most likely to subscribe. The bank can only contact 20% of customers, making accurate predictions critical for resource allocation. By identifying high-probability subscribers, the bank can concentrate marketing efforts on the most promising targets, reducing wasted contacts and improving campaign ROI.

From a theoretical standpoint, this problem involves binary classification under severe class imbalance, where only 11.3% of the 41,188 observations result in successful subscriptions. The challenge is to develop models that effectively discriminate between subscribers and non-subscribers while managing mixed data types, managing multicollinearity among economic indicators, and engineering temporal contact history variables. Additionally, the problem requires implementing down-sampling techniques to prevent model bias toward the majority "no" class and selecting evaluation metrics appropriate for imbalanced classification such as AUC, sensitivity, and specificity.

The remainder of this report is structured as follows: We begin with a comprehensive review of relevant literature pertaining to predictive modeling within the banking sector. Subsequently, we outline our methodology, detailing processes such as data preprocessing, feature engineering, model development, and validation techniques. The following section presents results that compare model performance and highlight principal predictors influencing subscription behavior. In conclusion, we offer recommendations aimed at optimizing customer targeting strategies and enhancing campaign efficiency.

III. Literature

Logistic regression, introduced by David Cox, is a method for binary classification problems due to its interpretability and probabilistic output. Linear Discriminant Analysis (LDA), developed by R.A. Fisher, provides an alternative classification approach by modeling class distributions and finding optimal linear separators.

Predictive models are theoretically effective in banking because they identify patterns in customer behavior that reduce uncertainty in decision-making. Vashi, Yadav, and Varde explain that these models use historical data to predict customer responses to marketing offers based on

demographic factors, economic conditions, and past interactions. This approach allows banks to target customers more effectively than random selection, optimizing marketing resources.

More recently, Peter et al. (2025) applied ensemble learning models to predict customer subscription in bank telemarketing campaigns, demonstrating that advanced machine learning techniques can achieve substantial improvements over traditional methods. Their study in Machine Learning with Applications emphasized the critical role of feature engineering and addressing class imbalance in achieving robust predictive performance.

In parallel with these developments, feature engineering for temporal marketing data, especially with respect to contact history variables that demonstrate non-linear associations with subscription outcomes, remains effective in models such as logistic regression and LDA. Nevertheless, the management of "unknown" categorical values and the approach to economic context variables require thoughtful attention to minimize multicollinearity and maintain predictive accuracy; this can be achieved within these models prior to introducing additional complexity.

IV. Methods

Logistic Regression and Linear Discriminant Analysis (LDA) were the models selected to address the binary classification problem of predicting bank term-deposit subscriptions. Logistic Regression was selected because of its high interpretability through odds ratios. Additionally, while Logistic Regression does assume that independent variables are not highly correlated, we decided to keep certain variables like *euribor3m,* and *nr.employed* despite their high VIF scores (54.328153 and 35.027662) because they were essential for predictive accuracy. Linear Discriminant Analysis was used as a second method to check if focusing on maximizing the distance between class means would yield a superior result.

The analysis was performed on a sample of 41,188 observations, providing a high degree of statistical power for the training and testing phases. The dataset did prove significant class imbalance, with only about 11% of customers subscribing to the term deposit. To prevent both models from developing a bias toward the majority of “No,” a down-sampling method was used while training the model. This approach balanced the training set by randomly reducing the number of majority class instances to match the minority class instances, forcing the model to learn the distinct characteristics of successful subscriptions.

To ensure the performance metrics did not result from overfitting to a specific subset of data, the following model validation techniques were used. A 5-fold cross-validation framework was used to ensure both models generalize well to unseen data. The bank data was split into an 80% training set and a 20% test set. Within the training set, the models were evaluated across five different folds using AUC (Area Under the ROC Curve), as the primary metric for optimization. However, a limitation of this study is that it relies on historical marketing data, and the model cannot account for sudden shifts in consumer sentiment or new banking regulations.

```
{r}
ctrl<-trainControl(method="cv",
                   number=5,
                   classProbs=TRUE,
                   summaryFunction=twoClassSummary,
                   sampling="down")
```

```
{r}
model_logit<-train(y~.,
                   data=train_data,
                   method="glm",
                   family="binomial",
                   metric="ROC",
                   trControl=ctrl)
model_lda<-train(y~., data=train_data, method="lda",
                 metric="ROC", trControl=ctrl)
```

V. Data

The bank marketing dataset contains 41,188 observations and 21 variables. The target variable *y* indicates whether a client subscribed to a term deposit ("yes" or "no"). The dataset exhibited severe class imbalance (88.7% "no", 11.3% "yes"), addressed using stratified sampling during train-test splitting.

The duration variable was excluded due to data leakage, it is only known after the call ends, making it unavailable for real-world predictions. Five economic context variables (employment variation rate, consumer price index, consumer confidence index, euribor 3-month rate, number of employees) exhibited high multicollinearity. To address this, we excluded *emp.var. rate* while retaining *cons.price.idx*, *cons.conf.idx*, *euribor3m*, and *nr. employed*, as *nr. employed* played a key role in the model despite the correlation.

Feature engineering created more informative predictors. The *age* variable was transformed into four categorical groups ("Young", "Middle-Age", "Mature", "Senior") to capture non-linear relationships. Campaign contacts (*campaign*) were categorized into intensity levels ("First_Contact", "Light_Followup", "Medium_Followup", "Heavy_Followup") to manage outliers. *Education* was converted to ordinal encoding (0-6) to preserve natural ordering (*education_level*). The *pdays* variable was transformed into a binary indicator (*pdays_binary*: "never_contacted" vs "contacted_before").
This showed that 86.8% of customers had never been previously contacted, validating the importance of this binary distinction for modeling. Additionally, *call_frequency_ratio* was engineered to capture contact intensity adjusted for recency.

Several variables contained "*unknown*" values, treated as valid categories except for *education_level*, where they were removed with minimal data loss.

The final dataset contains 14 predictors: 8 categorical (job, marital, housing, contact, poutcome, age_group, campaign_intensity, pdays_binary) and 5 numerical (cons.price.idx, cons.conf.idx, euribor3m, education_level, call_frequency_ratio). This reduction from 21 to 14 variables improves interpretability while maintaining predictive power.

VI. Results

The performance of the two models was evaluated on the 20% hold out test set. The results seemed almost identical; however, the Logistic Regression model outperformed the LDA model in terms of AUC. The outputs of the LDA model showed AUC to be 0.7646, sensitivity at 0.6648, and specificity at 0.7756. The Logistic Regression model output AUC at 0.7653, sensitivity at 0.6556, and specificity at 0.7858. As shown, the Logistic model has a slightly higher specificity than the LDA proving that the Logistic model is better at ensuring the bank does not market towards people who will say no. Moreover, the use of down-sampling resulted in a high sensitivity score, indicating that the model is highly effective at identifying true positives, which is the primary goal for the bank's marketing efficiency. The Logistic regression model also achieved a Lift score of 2.89x over a random strategy.

```
{r}
performance_comp<-data.frame(
  Metric=c("AUC", "Sensitivity", "Specificity"),
  Logistic=c(as.numeric(auc(roc_logit)),
             cm_logit$byClass["Sensitivity"],
```

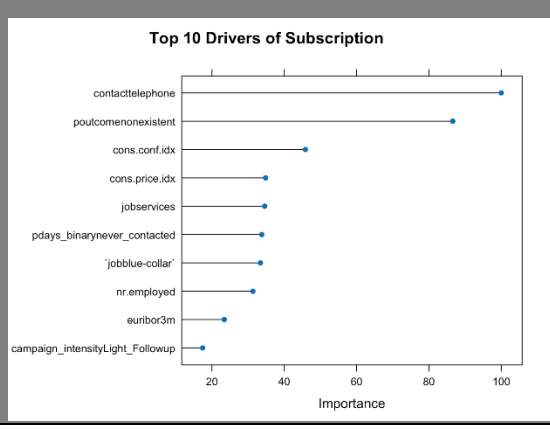
```
              cm_logit$byClass["Specificity"]),
  LDA=c(as.numeric(auc(roc_lda)),
        cm_lda$byClass["Sensitivity"],
        cm_lda$byClass["Specificity"])
)
cat("===Full Model Comparison===\n")
performance_comp[,-1]<-round(performance_comp[,-1],4)
print(performance_comp)
```

```
                Metric Logistic    LDA
                   AUC   0.7653 0.7646
Sensitivity Sensitivity   0.6556 0.6648
Specificity Specificity   0.7858 0.7756
```

After analyzing the Logistic Regression model's feature importance, we discovered that the method of contact and macroeconomic indicators are the strongest predictors of consumer behavior. The *contacttelephone* variable proved to be the top driver, suggesting that landline-based outreach significantly correlates with subscription likelihood in this demographic. Variables like the 3-month Euribor rate and Consumer Confidence Index also proved to be highly significant. This tells us that subscription rates are not purely a result of marketing but are tied to the economic environment as well. Lastly, the *poutcomeSuccess* and *pdays_binary* variables confirmed that existing subscriptions are a strong indicator of future sales.

VII. Insights

```
{r}
plot(varImp(model_logit), top=10, main="Top 10 Drivers of
Subscription")
```



The results of the Logistic Regression model provide meaningful insights into the key drivers of customer subscription behavior. Based on the model's variable importance output, three categories of predictors stand out: contact method, prior campaign history, and macroeconomic conditions.

First, the contact method was one of the strongest predictors of subscription. The variable *contacttelephone* showed high importance, indicating that customers reached via telephone were more likely to subscribe compared to other contact channels. This suggests that direct and personal outreach may increase customer engagement when promoting long-term financial products.

Second, prior engagement variables were highly influential. Customers, with a previous successful campaign outcome (*poutcomeSuccess*) demonstrated significantly higher likelihood of subscribing again. Similarly, the binary variable distinguishing whether a customer had been contacted before (*pdays_binary*) played an important predictive role. These findings support the importance of relationship-based marketing, where prior engagement increases responsiveness to future campaigns.

The engineered variable *call_frequency_ratio* also contributed to predictive performance. However, campaign intensity categories suggest diminishing returns from excessive follow-up. Customers in heavy follow-up categories did not exhibit proportionally higher subscription rates, indicating that increasing contact frequency alone does not guarantee improved outcomes.

Third, several macroeconomic indicators, including *euribor3m*, *cons.conf.idx*, and *nr.employed*, were among the most important predictors. This demonstrates that subscription decisions are influenced not only by marketing actions but also by broader economic conditions. Changes in interest rates and employment levels appear to shape customers' willingness to commit funds to term deposits. Notably, traditional demographic variables such as job category and marital status were less influential than expected. This suggests that behavioral history and economic context are stronger determinants of response than static demographic characteristics.

Overall, the findings indicate that successful targeting should prioritize prior engagement history and economic timing rather than relying heavily on demographic segmentation.

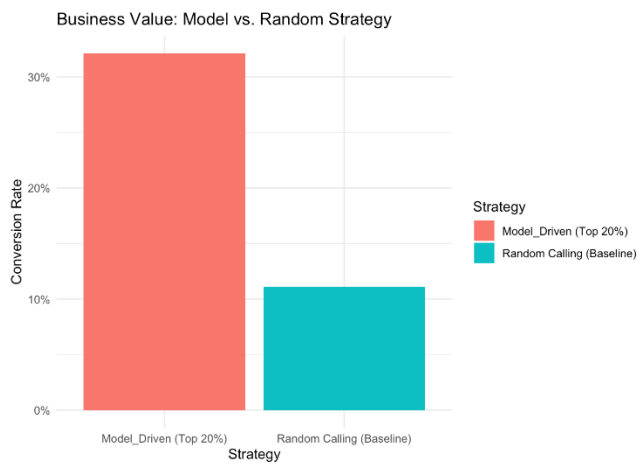
VIII. Strategy & Recommendations

Given the bank's constraint of contacting only 20% of customers, the optimal strategy is to rank customers based on predicted probabilities from the Logistic Regression model and select the top 20% highest-scoring individuals.

Using this model-driven approach, the conversion rate within the top 20% segment was 32.13%, compared to a baseline rate of 11.12% under random targeting. This corresponds to a Lift of 2.89, meaning the predictive model nearly triple the effectiveness of customer selection.

```
conv_top20<-mean(top_20_customers$y=="Yes")
conv_baseline<-mean(test_data$y=="Yes")
lift_score<-conv_top20/conv_baseline
```

```
{r}
ggplot(results_df, aes(x=Strategy, y=Conversion,
fill=Strategy))+
  geom_bar(stat="identity")+
  scale_y_continuous(labels=scales::percent)+
  theme_minimal()+
  labs(title="Business Value: Model vs. Random Strategy",
y="Conversion Rate")
```



From a business perspective, this improves marketing efficiency. Instead of allocating resources evenly across all customers, the bank can focus outreach efforts on those with the highest predicted likelihood of subscription, thereby maximizing return on investment.

Based on the findings, the bank should:

1. Prioritize customers with prior successful campaign outcomes.

2. Emphasize telephone outreach for high-probability segments.

3. Monitor macroeconomic indicators when planning campaign timing.

4. Avoid excessive follow-up intensity that does not proportionally increase conversion probability.

However, predictive targeting must be implemented responsibly. Although demographic variables were not dominant predictors in this model, the use of behavioral and socioeconomic indicators could indirectly disadvantage certain groups. The bank should periodically audit model outputs across age groups and other customer segments to ensure that targeting strategies do not unintentionally reinforce bias.

IX. Conclusion

This study aimed to improve the efficiency of bank marketing campaigns by developing predictive models for term-deposit subscriptions. Using Logistic Regression and Linear Discriminant Analysis on a dataset of 41,188 observations, both models demonstrated comparable performance, with Logistic Regression slightly outperforming LDA in terms of AUC and specificity. When applied under a realistic business constraint of contacting only 20% of customers, the model-driven strategy increased the conversion rate from 11.12% to 32.13%, achieving a Lift of 2.89 compared to random targeting.

Beyond predictive performance, the analysis revealed that prior campaign success, contact method, and macroeconomic conditions were the strongest drivers of subscription behavior. These findings suggest that marketing effectiveness depends more on engagement history and economic context than on static demographic characteristics. Overall, the results demonstrate that data-driven targeting can significantly enhance campaign efficiency while highlighting the importance of responsible implementation and continuous monitoring of model performance.

Resources

S., Qomariyah, N. N., Kazakov, D., & Fajar, A. N. (2025). Predict customer subscription on bank telemarketing campaign using ensemble learning. *Machine Learning with Applications, 19*, 100575.

Vashi, H., Yadav, J., & Varde, A. S. (n.d.). *Predictive analysis of bank marketing for financial decision support and smart economy*. Montclair State University.

X. Appendix

```
library(MASS) library(ggplot2) library(tidyverse)
library(corrplot) library(car) library(dplyr) library(caret)
library(pROC) data<-read.csv("~/Downloads/data analytics
application/bank-additional-full.csv", sep=";") glimpse(data)
Rows: 41,188 Columns: 21 $ age 56, 57, 37… $ job "housemaid… $
marital "married",… $ education "basic.4y"… $ default "no",
"unk… $ housing "no", "no"… $ loan "no", "no"… $ contact
"telephone… $ month "may", "ma… $ day_of_week "mon", "mo… $
duration 261, 149, … $ campaign 1, 1, 1, 1… $ pdays 999, 999, …
$ previous 0, 0, 0, 0… $ poutcome "nonexiste… $ emp.var.rate
1.1, 1.1, … $ cons.price.idx 93.994, 93… $ cons.conf.idx -36.4,
-36… $ euribor3m 4.857, 4.8… $ nr.employed 5191, 5191… $ y
"no", "no"… levels(data$Attrition_Flag) NULL dt_final<-data|>

  • mutate(
  • job=as.factor(job),

  • marital=as.factor(marital),

  • education=as.factor(education),

  • default=as.factor(default),

  • housing=as.factor(housing),

  • loan=as.factor(loan),

  • contact=as.factor(contact),

  • month=as.factor(month),

  • day_of_week=as.factor(day_of_week),

  • poutcome=as.factor(poutcome),

  • y=factor(y, levels=c("no", "yes"), labels=c("No", "Yes"))

  • )|>
```

```
• mutate(
• call_frequency_ratio = case_when(

•  pdays==999~campaign / 1,

•  pdays==0~(campaign+previous)/1,

•  TRUE~(campaign+previous)/(pdays+1)

• ),

• pdays_binary=factor(ifelse(pdays==999, "never_contacted",
  "contacted_before"))

• )|>
• mutate(
• age_group=case_when(

•  age < 30 ~ "Young",

•  age < 45 ~ "Middle-Age",

•  age < 60 ~ "Mature",

•  TRUE ~ "Senior"

• ),

• age_group=as.factor(age_group)

• )|>
• mutate(
• campaign_intensity=case_when(

•  campaign==1 ~ "First_Contact",

•  campaign <=3~ "Light_Followup",

•  campaign <=7 ~ "Medium_Followup",

•  TRUE ~ "Heavy_Followup"
```

```
  • ),

  • campaign_intensity = as.factor(campaign_intensity)

  • )|>
  • mutate(
  • education_level = case_when(

  •  education =="illiterate"~0,

  •  education == "basic.4y"~1,

  •  education=="basic.6y"~2,

  •  education=="basic.9y"~3,

  •  education=="high.school"~4,

  •  education=="professional.course"~5,

  •  education=="university.degree"~6,

  •  TRUE~NA_real_

  • )

  • )|>
  • dplyr::select(-duration, -emp.var.rate, -age, -campaign, -
    education, -pdays, -loan, -month, -day_of_week, -pdays, -
    previous, -default )|>
  • na.omit()

set.seed(123) trainIndex<-createDataPartition(dt_final$y, p=.8,
list=FALSE) train_data<- dt_final[trainIndex, ] test_data<-
dt_final[-trainIndex, ] ctrl<-trainControl(method="cv",

  •               number=5,

  •               classProbs=TRUE,

  •               summaryFunction=twoClassSummary,

  •               sampling="down")
```

```
model_logit<-train(y~.,

   •                data=train_data,

   •                method="glm",

   •                family="binomial",

   •                metric="ROC",

   •                trControl=ctrl)


vif(model_logit$finalModel) jobblue-collar 2.087930
jobentrepreneur 1.111932 jobhousemaid 1.159707 jobmanagement
1.206265 jobretired 1.791925 jobself-employed 1.098821
jobservices 1.326558 jobstudent 1.208642 jobtechnician 1.358449
jobunemployed 1.091912 jobunknown 1.023915 maritalmarried
2.724329 maritalsingle 3.048856 maritalunknown 1.026481
housingunknown 1.032153 housingyes 1.043651 contacttelephone
1.606436 poutcomenonexistent 1.448257 poutcomesuccess 8.156922
cons.price.idx 6.524123 cons.conf.idx 2.834936 euribor3m
54.328153 nr.employed 35.027662 call_frequency_ratio 5.478768
pdays_binarynever_contacted 8.499524 age_groupMiddle-Age
1.605473 age_groupSenior 1.626034 age_groupYoung 1.832470
campaign_intensityHeavy_Followup 4.339191
campaign_intensityLight_Followup 1.578635
campaign_intensityMedium_Followup 2.979536 education_level
1.750126 model_lda<-train(y~., data=train_data, method="lda",

   •              metric="ROC", trControl=ctrl)


test_probs<-predict(model_logit, newdata=test_data,
type="prob") prob_logit<- predict(model_logit,
newdata=test_data, type="prob")$Yes prob_lda<-
predict(model_lda, newdata=test_data, type="prob")$Yes
roc_logit<-roc(test_data$y, prob_logit, quiet=TRUE) roc_lda <-
roc(test_data$y, prob_lda, quiet=TRUE) cat("===AUC
Comparison===\n") ===AUC Comparison=== cat("Logistic Regression
AUC:", auc(roc_logit), "\n") Logistic Regression AUC: 0.7652747
cat("LDA AUC:", auc(roc_lda), "\n") LDA AUC: 0.7646249

pred_logit<-predict(model_logit, newdata=test_data) pred_lda<-
predict(model_lda, newdata=test_data) cm_logit<-
confusionMatrix(pred_logit, test_data$y, positive="Yes")
```

```
cm_lda<-confusionMatrix(pred_lda, test_data$y, positive="Yes")
performance_comp<-data.frame(

  • Metric=c("AUC", "Sensitivity", "Specificity"),
  • Logistic=c(as.numeric(auc(roc_logit)),
  •         cm_logit$byClass["Sensitivity"],

  •         cm_logit$byClass["Specificity"]),

  • LDA=c(as.numeric(auc(roc_lda)),
  •    cm_lda$byClass["Sensitivity"],

  •    cm_lda$byClass["Specificity"])

  • )

cat("===Full Model Comparison===\n") ===Full Model
Comparison=== performance_comp[,-1]<-round(performance_comp[,-
1],4) print(performance_comp) plot(varImp(model_logit), top=10,
main="Top 10 Drivers of Subscription") strategic_list<-
test_data|>

  • mutate(Prob_yes=test_probs$Yes)|>
  • arrange(desc(Prob_yes))

top_20_cutoff <- floor (0.20*nrow(strategic_list))
top_20_customers<-strategic_list[1:top_20_cutoff, ]
conv_top20<-mean(top_20_customers$y=="Yes") conv_baseline<-
mean(test_data$y=="Yes") lift_score<-conv_top20/conv_baseline
results_df<-data.frame(

  • Strategy=c("Random Calling (Baseline)", "Model_Driven (Top
    20%)"),
  • Conversion=c(conv_baseline, conv_top20)
  • )

ggplot(results_df, aes(x=Strategy, y=Conversion,
fill=Strategy))+

  • geom_bar(stat="identity")+
  • scale_y_continuous(labels=scales::percent)+
  • theme_minimal()+
  • labs(title="Business Value: Model vs. Random Strategy",
    y="Conversion Rate")
```

```
cat("==Final Model Performance===\n") ==Final Model
Performance=== cat("Conversion in Top 20% Target Group:",
round(conv_top20 100,2), "%\n") Conversion in Top 20% Target
Group: 32.13 % cat("Conversion in Random Targeting
(Baseline):", round(conv_baseline100,2), "%\n") Conversion in
Random Targeting (Baseline): 11.12 % cat("Model Lift",
round(lift_score, 2), "x better than random\n") Model Lift 2.89
x better than random
```